Midterm Review with Binary Effort and Action Feedback

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Abstract

This note studies a two-period moral-hazard problem with binary effort and additive standard normal noise. Shirking is costless and produces zero expected progress, while working costs $c > 0$ and raises expected progress by $\mu > 0$. Monetary units are normalized so that project completion is worth one to the principal. First-period progress is privately observed by the principal at a midterm review, but the principal cannot commit to report that progress truthfully. We impose the economically central restriction that first-period work is uniquely first-best. We first characterize the first best and complete pooling. We then study binary action feedback: after any noncompletion outcome, the principal recommends either work or shirk in the second period. The two shirk tails are pooled, and the principal reports only information that changes the desired continuation action. Under normal noise, the principal's reporting payoff is single-peaked, so any bounded work recommendation region is an interval. Boundary indifference eliminates the reporting wages and reduces every fixed-interval contract to a one-dimensional, piecewise-linear payment problem. We characterize both the first-period work and first-period shirk branches, discuss one-sided feedback, and formulate an unrestricted three-message benchmark. We prove that, at and immediately below the diagonal $\theta = \mu$, complete pooling with work in both periods is optimal for sufficiently small effort cost. In this region contractual feedback is strictly coarser than the first best: the first best recommends shirking (giving up) after a sufficiently low realization, whereas the optimal contract pools all noncompletion outcomes and induces continuation work everywhere. Numerical diagnostics indicate that binary action feedback is optimal on a nonempty intermediate-cost region, while a third continuation message adds essentially no value whenever feedback is globally optimal.

1 Environment

There are two periods. In each period the agent chooses effort $a \in \{0, 1\}$. Shirking, $a = 0$, is costless and produces zero expected progress. Working, $a = 1$, costs $c > 0$ and raises expected progress by $\mu > 0$. Conditional on effort a, progress is

$$Z = a\mu + \varepsilon, \qquad \varepsilon \sim N(0, 1), \tag{1}$$

with independent shocks across periods. Let Φ and ϕ denote the standard normal CDF and density. The first-period density under effort a is

$$g_a(z) = \phi(z - a\mu). \tag{2}$$

The likelihood ratio

$$\lambda(z) := \frac{g_1(z)}{g_0(z)} = \exp\left\{\mu z - \frac{\mu^2}{2}\right\} \tag{3}$$

is strictly increasing.

The project succeeds immediately if

$$Z_1 \geq \theta. \tag{4}$$

If $Z_1 = z < \theta$, a second period occurs, and the project succeeds if

$$z + Z_2 \geq \theta. \tag{5}$$

Success is publicly observed. Monetary units are normalized so that the principal receives one upon completion and zero otherwise. The effort cost c and all contractual payments are therefore measured in units of the completion prize.

For a first-period realization $z < \theta$, the continuation success probability under shirking is

$$p(z) := \Phi(z - \theta), \tag{6}$$

and the continuation success probability under work is

$$q(z) := \Phi(z + \mu - \theta). \tag{7}$$

The incremental productivity of second-period work is

$$D(z) := q(z) - p(z). \tag{8}$$

Define

$$z_m := \theta - \frac{\mu}{2}. \tag{9}$$

Lemma 1 (Single-peaked continuation productivity)**.** *The function D is symmetric around z_m, strictly increasing on $(-\infty, z_m)$, and strictly decreasing on (z_m, ∞). Its maximum is*

$$D_{\max} = 2\Phi\left(\frac{\mu}{2}\right) - 1. \tag{10}$$

Moreover, $D(z) \to 0$ as $z \to \pm\infty$.

Proof. For $z = z_m + r$,

$$D(z_m + r) = \Phi\left(r + \frac{\mu}{2}\right) - \Phi\left(r - \frac{\mu}{2}\right), \tag{11}$$

which is even in r. Differentiation gives

$$D'(z_m + r) = \phi\left(r + \frac{\mu}{2}\right) - \phi\left(r - \frac{\mu}{2}\right). \tag{12}$$

The derivative is positive for $r < 0$ and negative for $r > 0$. □

2 The first best

After observing $z < \theta$, second-period work is first-best if and only if

$$D(z) \geq c. \tag{13}$$

The shape of the first-best continuation policy therefore depends on c.

Proposition 1 (First-best continuation regimes)**.** *The first-best second-period action has the following form.*

(i) *If $0 < c < D(\theta)$, there is a unique lower cutoff $\ell^{\mathrm{FB}} < z_m$ satisfying*

$$D(\ell^{\mathrm{FB}}) = c, \tag{14}$$

and work is optimal for

$$\ell^{\mathrm{FB}} \leq z < \theta. \tag{15}$$

(ii) *If $D(\theta) < c < D_{\max}$, there are unique cutoffs*

$$\ell^{\mathrm{FB}} < z_m < h^{\mathrm{FB}} < \theta \tag{16}$$

satisfying

$$D(\ell^{\mathrm{FB}}) = D(h^{\mathrm{FB}}) = c, \tag{17}$$

and work is optimal precisely on $[\ell^{\mathrm{FB}}, h^{\mathrm{FB}}]$.

(iii) *If $c \geq D_{\max}$, second-period work is never optimal.*

At the boundary values the corresponding inequalities may be taken weakly.

Proof. The result follows directly from (**??**) and Lemma **??**. □

Define the first-best value conditional on first-period progress by

$$W^{\mathrm{FB}}(z) = \begin{cases} 1, & z \geq \theta, \\ p(z) + [D(z) - c]_+, & z < \theta. \end{cases} \tag{18}$$

The gross gain from working rather than shirking in period one is

$$\Gamma^{\mathrm{FB}}(\mu, \theta, c) := \int_{-\infty}^{\infty} \big[g_1(z) - g_0(z)\big] W^{\mathrm{FB}}(z)\, dz. \tag{19}$$

Assumption 1 (Social value of pre-review effort)**.** *The primitives satisfy*

$$\Gamma^{\mathrm{FB}}(\mu, \theta, c) > c. \tag{20}$$

Proposition 2 (First-period effort)**.** *Under Assumption **??**, first-period work is uniquely first-best. Moreover,*

$$c < D_{\max}. \tag{21}$$

Consequently, the first-best continuation policy is described by part (i) *or part* (ii) *of Proposition **??**.*

Proof. Expected first-best surplus under first-period effort a is

$$\int g_a(z) W^{\mathrm{FB}}(z)\, dz - ac. \tag{22}$$

The difference between work and shirk is $\Gamma^{\mathrm{FB}} - c$, which is strictly positive by assumption.

For the second statement, W^{FB} takes values in $[0, 1]$. The difference between its expectations under g_1 and g_0 is therefore bounded by the total-variation distance between $N(\mu, 1)$ and $N(0, 1)$, which equals

$$2\Phi\left(\frac{\mu}{2}\right) - 1 = D_{\max}. \tag{23}$$

Hence $\Gamma^{\mathrm{FB}} \leq D_{\max}$. Since $\Gamma^{\mathrm{FB}} > c$, we obtain $c < D_{\max}$. □

Remark 1. *The maintained assumption concerns only first-period efficiency. It does not impose a particular continuation regime. In particular, both a one-sided first-best work region and a strict middle work band remain possible.*

3 Contracts and double deviations

The principal privately observes first-period progress whenever the project has not already succeeded. A continuation message m specifies a failure wage $w_m \geq 0$ and a success bonus $b_m \geq 0$. If the agent receives message m, shirks in period two, and the project succeeds, he receives $w_m + b_m$; if the project fails, he receives w_m. If he works in period two, the success probability rises by $D(z)$ and he incurs cost c.

An immediate-success payment $R \geq 0$ may be paid when $Z_1 \geq \theta$. For a measurable report region $I_m \subset (-\infty, \theta)$ and first-period action a, define

$$\pi_m^a := \int_{I_m} g_a(z)\,dz, \qquad Q_m^a := \int_{I_m} g_a(z)p(z)\,dz, \qquad G_m^a := \int_{I_m} g_a(z)D(z)\,dz. \tag{24}$$

Conditional continuation work after report m is optimal if and only if

$$b_m G_m^a \geq c\pi_m^a. \tag{25}$$

The agent's continuation utility after first-period action a, allowing him to reoptimize his second-period action after every report, is

$$C_a = \sum_m \left\{ w_m \pi_m^a + b_m Q_m^a + [b_m G_m^a - c\pi_m^a]_+ \right\}. \tag{26}$$

Thus the positive-part formula incorporates every double deviation in a single expression.

If first-period work is intended, obedience requires

$$R(P_1 - P_0) + C_1 - C_0 \geq c. \tag{27}$$

If first-period shirking is intended, obedience requires the reverse weak inequality.

4 Complete pooling

Under complete pooling, every noncompletion outcome receives the same continuation contract (w, b). Define

$$P_a := \Pr_a\{Z_1 \geq \theta\}, \qquad N_a := 1 - P_a, \qquad \Delta P := P_1 - P_0 > 0, \tag{28}$$

$$Q_a := \int_{-\infty}^{\theta} g_a(z)p(z)\,dz, \qquad G_a := \int_{-\infty}^{\theta} g_a(z)D(z)\,dz, \qquad A_a := Q_a + G_a. \tag{29}$$

The agent's optimized utility after first-period action a is

$$U_a(R, w, b) = RP_a + wN_a + bQ_a + [bG_a - cN_a]_+ - ac. \tag{30}$$

Continuation work after action a occurs if and only if

$$b \geq \beta_a^{\text{pool}} := \frac{cN_a}{G_a}. \tag{31}$$

Let the first letter in (X, Y) denote the first-period action and the second the pooled continuation action, where W denotes work and S denotes shirking.

Proposition 3 (Optimal pooled contracts)**.** *The four pooled effort allocations have the following least-cost implementations.*

(i) *For* (S,S),

$$(R,w,b)=(0,0,0), \tag{32}$$

and

$$V_{SS}^{\text{pool}}=P_0+Q_0. \tag{33}$$

(ii) *For* (W,S),

$$(R,w,b)=\left(\frac{c}{\Delta P},0,0\right), \tag{34}$$

and

$$V_{WS}^{\text{pool}}=P_1+Q_1-\frac{P_1c}{\Delta P}. \tag{35}$$

(iii) *For* (S,W),

$$R=0, \qquad b=\beta_0^{\text{pool}}. \tag{36}$$

The least failure wage is

$$w_{SW}^{\text{pool}}=\begin{cases}0, & \beta_0^{\text{pool}}<\beta_1^{\text{pool}},\\ \dfrac{\left[\beta_0^{\text{pool}}(A_1-A_0)-c(1-\Delta P)\right]_+}{\Delta P}, & \beta_0^{\text{pool}}\geq\beta_1^{\text{pool}}.\end{cases} \tag{37}$$

The principal's value is

$$V_{SW}^{\text{pool}}=P_0+A_0-\beta_0^{\text{pool}}A_0-N_0w_{SW}^{\text{pool}}. \tag{38}$$

(iv) *For* (W,W), *set* $w=0$. *For a proposed bonus* b, *define*

$$H_S^{WW}(b):=c(1+N_1)-b(A_1-Q_0), \tag{39}$$

when a first-period shirker would also shirk in period two, and

$$H_W^{WW}(b):=c(1-\Delta P)-b(A_1-A_0), \tag{40}$$

when that deviator would work. These expressions agree at $b=\beta_0^{\text{pool}}$. *The least immediate-success reward is*

$$R^{WW}(b)=\frac{\left[H_\delta^{WW}(b)\right]_+}{\Delta P}. \tag{41}$$

If

$$\beta_0^{\text{pool}}\leq\beta_1^{\text{pool}}, \tag{42}$$

or

$$H_S^{WW}(\beta_1^{\text{pool}})\leq 0, \tag{43}$$

or

$$P_0A_1-P_1Q_0\leq 0, \tag{44}$$

then

$$b_{WW}^{\text{pool}}=\beta_1^{\text{pool}}. \tag{45}$$

Otherwise define

$$b_R^{\text{pool}}:=\frac{c(1+N_1)}{A_1-Q_0}, \tag{46}$$

and set

$$b_{WW}^{\text{pool}} = \min\left\{b_R^{\text{pool}}, \beta_0^{\text{pool}}\right\}. \tag{47}$$

The value is

$$V_{WW}^{\text{pool}} = P_1 + A_1 - b_{WW}^{\text{pool}} A_1 - P_1 R^{WW}(b_{WW}^{\text{pool}}). \tag{48}$$

Consequently,

$$V^{\text{pool}} = \max\left\{V_{SS}^{\text{pool}}, V_{WS}^{\text{pool}}, V_{SW}^{\text{pool}}, V_{WW}^{\text{pool}}\right\}. \tag{49}$$

Proof. The first two cases follow immediately from (**??**) and the relative efficiency of immediate success for inducing first-period work. For (S, W), an immediate-success payment encourages the unwanted first-period action, so $R = 0$. The principal's value is piecewise linear and decreasing in the continuation bonus, so the on-path continuation constraint binds at $b = \beta_0^{\text{pool}}$; the wage in (**??**) is the least amount needed to deter first-period work.

For (W, W), a failure wage discourages the intended first-period action, so $w = 0$. Above the point at which a deviating first-period shirker also works in period two, the value is strictly decreasing in b. Below that point, raising b is useful only if it reduces the immediate-success reward faster than it raises on-path continuation payments. The stated cases are the resulting piecewise-linear vertex conditions. □

Remark 2. *Under Assumption* **??**, (S, S) *is a strict downward distortion of first-period effort. It remains a useful rent-free benchmark, but it is never first-best.*

5 Binary action feedback

After any noncompletion outcome, the principal sends one of two messages:

$$S: \text{ shirk in period two,} \qquad W: \text{ work in period two.} \tag{50}$$

The corresponding contracts are (w_S, b_S) and (w_W, b_W). We focus on the regular branch

$$0 \le b_S \le 1, \qquad 0 \le b_W \le 1, \tag{51}$$

so that the principal's continuation payoff is weakly increasing in success under either report. All numerical optima reported below lie on this branch.

If the agent shirks after S and works after W, the principal's continuation payoffs are

$$V_S(z) = (1 - b_S)p(z) - w_S, \tag{52}$$

$$V_W(z) = (1 - b_W)q(z) - w_W. \tag{53}$$

The reporting-payoff difference is

$$A(z) := V_W(z) - V_S(z). \tag{54}$$

Lemma 2 (Single-peaked reporting payoff)**.** *On the regular branch, A is single-peaked. Hence the set of realizations for which reporting W is sequentially optimal is an interval, possibly empty, one-sided, or bounded.*

Proof. Differentiation gives

$$A'(z) = (1 - b_W)\phi(z + \mu - \theta) - (1 - b_S)\phi(z - \theta). \tag{55}$$

The ratio

$$\frac{\phi(z + \mu - \theta)}{\phi(z - \theta)} = \exp\left\{-\mu(z - \theta) - \frac{\mu^2}{2}\right\} \tag{56}$$

is strictly decreasing, so A' crosses zero at most once. □

5.1 Bounded work intervals

Fix

$$-\infty < \ell < h < \theta \tag{57}$$

and consider the intended reporting rule

$$W \quad \Longleftrightarrow \quad z \in [\ell, h]. \tag{58}$$

Boundary indifference requires

$$V_W(\ell) = V_S(\ell), \qquad V_W(h) = V_S(h). \tag{59}$$

Define

$$\alpha(\ell, h) := \frac{p(h) - p(\ell)}{q(h) - q(\ell)}. \tag{60}$$

Subtracting the two boundary conditions gives

$$1 - b_W = \alpha(\ell, h)(1 - b_S). \tag{61}$$

Parameterize the principal's success shares by

$$1 - b_S = r, \qquad 1 - b_W = \alpha r. \tag{62}$$

Thus

$$b_S = 1 - r, \qquad b_W = 1 - \alpha r. \tag{63}$$

Bonus nonnegativity requires

$$0 \leq r \leq \min\left\{1, \frac{1}{\alpha}\right\}. \tag{64}$$

Define

$$\kappa(\ell, h) := \alpha q(\ell) - p(\ell). \tag{65}$$

The ratio $p(z)/q(z)$ is strictly increasing, which implies $\kappa > 0$. The least nonnegative wage pair satisfying boundary indifference is therefore

$$w_S = 0, \qquad w_W = r\kappa. \tag{66}$$

Any other wage pair adds a common continuation wage $t \geq 0$:

$$w_S = t, \qquad w_W = t + r\kappa. \tag{67}$$

Lemma 3 (Binary reporting envelope)**.** *Fix $\ell < h < \theta$. Every regular binary contract implementing* (**??**) *can be written as* (**??**) *and* (**??**) *for some*

$$0 < r \leq \min\left\{1, \frac{1}{\alpha}\right\}, \qquad t \geq 0. \tag{68}$$

Conversely, every such contract satisfies all global principal reporting constraints.

Proof. Necessity follows from the two boundary conditions. Under the stated parameterization,

$$V_W(z) - V_S(z) = r\left[\alpha q(z) - p(z) - \kappa\right]. \tag{69}$$

The bracketed function is single-peaked and equals zero at both ℓ and h. It is therefore positive on (ℓ, h) and negative outside that interval. The common wage cancels from the reporting comparison. □

5.2 Continuation obedience and double deviations

For $a \in \{0,1\}$, define

$$\pi_W^a := \int_\ell^h g_a(z)\,dz, \qquad \pi_S^a := N_a - \pi_W^a, \tag{70}$$

$$Q_W^a := \int_\ell^h g_a(z)p(z)\,dz, \qquad Q_S^a := Q_a - Q_W^a, \tag{71}$$

$$G_W^a := \int_\ell^h g_a(z)D(z)\,dz, \qquad G_S^a := G_a - G_W^a. \tag{72}$$

Whenever $\pi_j^a > 0$, write

$$\overline{D}_j^a := \frac{G_j^a}{\pi_j^a}, \qquad j \in \{S, W\}. \tag{73}$$

Suppose first-period effort $x \in \{0,1\}$ is intended. Continuation obedience requires

$$(1-r)\overline{D}_S^x \le c, \tag{74}$$

$$(1-\alpha r)\overline{D}_W^x \ge c. \tag{75}$$

Define

$$\underline{r}_x(\ell,h) := \max\left\{0, 1 - \frac{c}{\overline{D}_S^x}\right\}, \tag{76}$$

$$\overline{r}_x(\ell,h) := \min\left\{1, \frac{1}{\alpha}, \frac{1 - c/\overline{D}_W^x}{\alpha}\right\}. \tag{77}$$

The interval is feasible on branch x exactly when

$$\overline{D}_W^x \ge c \tag{78}$$

and

$$\underline{r}_x(\ell,h) \le \overline{r}_x(\ell,h). \tag{79}$$

The feasible slope set is

$$\mathcal{R}_x(\ell,h) := [\underline{r}_x(\ell,h), \overline{r}_x(\ell,h)]. \tag{80}$$

Excluding the common wage t, continuation utility following first-period action a is

$$\begin{aligned}\widehat{C}_a(r) = r\kappa\pi_W^a + (1-r)Q_S^a + (1-\alpha r)Q_W^a \\ + [(1-r)G_S^a - c\pi_S^a]_+ + [(1-\alpha r)G_W^a - c\pi_W^a]_+ .\end{aligned} \tag{81}$$

A common wage adds tN_a.

5.3 The first-period work branch

A common continuation wage weakens first-period work incentives because $N_0 > N_1$. Hence an optimal work-branch contract has

$$t_1^* = 0. \tag{82}$$

For a fixed r, the least immediate-success payment is

$$R_1(r) = \frac{\left[c - \widehat{C}_1(r) + \widehat{C}_0(r)\right]_+}{\Delta P}. \tag{83}$$

On-path success probability is

$$\Sigma_1(\ell,h) := P_1 + Q_1 + G_W^1. \tag{84}$$

Expected continuation transfers are

$$\begin{aligned}\widehat{K}_1(r) &= r\kappa\pi_W^1 + (1-r)Q_S^1 \\ &\quad + (1-\alpha r)(Q_W^1 + G_W^1).\end{aligned} \tag{85}$$

The fixed-interval value is

$$V_1^{\mathrm{WS}}(\ell,h;r) = \Sigma_1(\ell,h) - \widehat{K}_1(r) - P_1R_1(r). \tag{86}$$

Proposition 4 (Fixed-interval work branch)**.** *For every feasible interval (ℓ,h),*

$$V_1^{\mathrm{WS}}(\ell,h) = \max_{r\in\mathcal{R}_1(\ell,h)} V_1^{\mathrm{WS}}(\ell,h;r). \tag{87}$$

The objective is continuous and piecewise affine in r. An optimum may be chosen from the finite set consisting of

(i) *the endpoints of $\mathcal{R}_1(\ell,h)$;*

(ii) *the off-path continuation-action thresholds*

$$r = 1 - \frac{c}{\overline{D}_S^0}, \qquad r = \frac{1-c/\overline{D}_W^0}{\alpha}; \tag{88}$$

(iii) *the roots, within each off-path action regime, of*

$$c - \widehat{C}_1(r) + \widehat{C}_0(r) = 0. \tag{89}$$

Proof. The positive-part terms in (**??**) are maxima of affine functions, so the first-period incentive deficit is piecewise affine in r. The remaining transfer and success terms are affine. Therefore the value is piecewise affine, and every maximum occurs at an endpoint or kink. □

5.4 The first-period shirk branch

An immediate-success reward encourages the unwanted first-period action, so an optimal shirk-branch contract has

$$R_0^* = 0. \tag{90}$$

For fixed r, the least common wage deterring first-period work is

$$t_0(r) = \frac{\left[\widehat{C}_1(r) - \widehat{C}_0(r) - c\right]_+}{\Delta P}. \tag{91}$$

On-path success probability is

$$\Sigma_0(\ell,h) := P_0 + Q_0 + G_W^0. \tag{92}$$

Expected continuation transfers excluding the common wage are

$$\begin{aligned}\widehat{K}_0(r) &= r\kappa\pi_W^0 + (1-r)Q_S^0 \\ &\quad + (1-\alpha r)(Q_W^0 + G_W^0).\end{aligned} \tag{93}$$

Thus

$$V_0^{\mathrm{WS}}(\ell,h;r) = \Sigma_0(\ell,h) - \widehat{K}_0(r) - N_0t_0(r). \tag{94}$$

Proposition 5 (Fixed-interval shirk branch)**.** *For every feasible interval* (ℓ, h)*,*

$$V_0^{\mathrm{WS}}(\ell, h) = \max_{r \in \mathcal{R}_0(\ell,h)} V_0^{\mathrm{WS}}(\ell, h; r). \tag{95}$$

The objective is continuous and piecewise affine in r*. An optimum may be chosen from the finite set consisting of*

(i) *the endpoints of* $\mathcal{R}_0(\ell, h)$*;*

(ii) *the off-path continuation-action thresholds*

$$r = 1 - \frac{c}{\overline{D}_S^1}, \qquad r = \frac{1 - c/\overline{D}_W^1}{\alpha}; \tag{96}$$

(iii) *the roots, within each off-path action regime, of*

$$\widehat{C}_1(r) - \widehat{C}_0(r) - c = 0. \tag{97}$$

Proof. The proof is identical to that of Proposition **??**, with the common continuation wage replacing the immediate-success payment. ☐

5.5 One-sided action feedback

The numerically important special case recommends work after sufficiently high noncompletion progress:

$$W \iff \ell \le z < \theta. \tag{98}$$

Because there is no report at $z = \theta$, the upper reporting constraint need not bind. Define

$$\gamma := \frac{1 - b_W}{1 - b_S}, \tag{99}$$

and

$$\underline{\gamma}(\ell) := \max\left\{ \frac{p(\ell)}{q(\ell)}, \frac{p(\theta) - p(\ell)}{q(\theta) - q(\ell)} \right\}. \tag{100}$$

Lemma 4 (One-sided reporting obedience)**.** *Fix* $\ell < \theta$*. A regular contract with*

$$1 - b_S = r, \qquad 1 - b_W = \gamma r \tag{101}$$

implements (**??**) *if and only if*

$$\gamma \ge \underline{\gamma}(\ell) \tag{102}$$

and the wage difference satisfies

$$w_W - w_S = r\big[\gamma q(\ell) - p(\ell)\big]. \tag{103}$$

The least nonnegative wages set $w_S = 0$ *and choose* w_W *equal to the right-hand side.*

Proof. The reporting difference divided by r is

$$\gamma q(z) - p(z) - \big[\gamma q(\ell) - p(\ell)\big]. \tag{104}$$

It is single-peaked and vanishes at ℓ. The first term in (**??**) ensures that the payoff difference is nonpositive in the far lower tail; the second ensures that it remains nonnegative through the upper endpoint θ. ☐

For fixed (ℓ, γ), the continuation and first-period incentive analysis is identical to the bounded-interval case after replacing α by γ and using $W = [\ell, \theta)$. The fixed-partition payment problem is therefore again a finite piecewise-linear vertex problem.

Define

$$V_x^{\mathrm{WS,bd}} := \sup_{\ell < h < \theta} V_x^{\mathrm{WS}}(\ell, h), \tag{105}$$

$$V_x^{\mathrm{WS,up}} := \sup_{\substack{\ell < \theta, \\ \gamma \geq \underline{\gamma}(\ell)}} V_x^{\mathrm{WS}}(\ell, \gamma), \tag{106}$$

and

$$V^{\mathrm{WS}} := \max_{x \in \{0,1\}} \left\{ V_x^{\mathrm{WS,bd}}, V_x^{\mathrm{WS,up}} \right\}. \tag{107}$$

6 Unrestricted three-message benchmark

For comparison, consider three continuation reports L, M, H associated with low, middle, and high progress regions

$$I_L = (-\infty, \ell), \qquad I_M = [\ell, h], \qquad I_H = (h, \theta). \tag{108}$$

The intended continuation action is shirk after L and H, and work after M. The principal's continuation payoffs are

$$V_L(z) = (1 - b_L)p(z) - w_L, \tag{109}$$

$$V_M(z) = (1 - b_M)q(z) - w_M, \tag{110}$$

$$V_H(z) = (1 - b_H)p(z) - w_H. \tag{111}$$

Define

$$A_L(z) := V_M(z) - V_L(z), \qquad A_H(z) := V_M(z) - V_H(z), \tag{112}$$

$$T(z) := V_H(z) - V_L(z). \tag{113}$$

Lemma 5 (Finite reporting conditions)**.** *Suppose the bonuses lie on the regular branch and $b_L > b_H$. The intended $L/M/H$ reporting rule is globally sequentially rational if and only if*

$$A_L(\ell) = 0, \qquad A_H(h) = 0, \tag{114}$$

$$A_L'(\ell) \geq 0, \qquad A_H'(h) \leq 0, \tag{115}$$

and

$$A_H(\ell) \geq 0, \qquad A_L(h) \geq 0. \tag{116}$$

Proof. Each of A_L and A_H is single-peaked by the normal-density-ratio argument in Lemma **??**. The difference

$$T(z) = (b_L - b_H)p(z) - (w_H - w_L) \tag{117}$$

is strictly increasing when $b_L > b_H$. The boundary, derivative, and cross conditions are therefore necessary and sufficient for L to dominate below ℓ, M to dominate between the cutoffs, and H to dominate above h. $\square$

For fixed (ℓ, h) and a specified continuation-action profile after a deviation in first-period effort, boundary indifference eliminates two wage differences, limited liability determines the least common intercept, and all remaining continuation and first-period constraints are linear. Hence the fixed-cutoff problem is a finite collection of linear programs. Denote the resulting optimized values by

$$V_x^{\mathrm{LMH}}(\ell, h), \qquad x \in \{0,1\}, \tag{118}$$

and define

$$V^{\mathrm{LMH}} := \max_{x\in\{0,1\}} \sup_{\ell<h<\theta} V_x^{\mathrm{LMH}}(\ell, h). \tag{119}$$

The binary action-feedback class is nested in this benchmark by assigning identical contracts to L and H.

7 Low-cost pooling and strict coarsening

The next result identifies a region in which the optimal contractual feedback is strictly coarser than the first-best continuation policy. The restriction $\theta \le \mu$ is sufficient rather than necessary. Its role is to make the lower-tail comparison uniform: a very low first-period realization is no more likely under first-period shirking than the boundary success probability that must be used to separate that tail.

Fix constants

$$0 < \underline{\mu} < \overline{\mu}, \qquad 0 < \varepsilon < \underline{\mu}, \tag{120}$$

and define the compact strip

$$\mathcal{K} := \left\{(\mu, \theta) : \underline{\mu} \le \mu \le \overline{\mu}, \quad 0 \le \mu - \theta \le \varepsilon\right\}. \tag{121}$$

Thus $\mathcal{K}$ lies at and immediately below the diagonal $\theta = \mu$.

Proposition 6 (Low-cost pooling and strict coarsening)**.** *There exists $\bar{c} > 0$ such that, for every $(\mu, \theta) \in \mathcal{K}$ and every $c \in (0, \bar{c})$, the following statements hold.*

(i) *First-period work is uniquely first-best. There is a unique cutoff $\ell^{\mathrm{FB}}(c) < \theta - \mu/2$ satisfying*

$$D\big(\ell^{\mathrm{FB}}(c)\big) = c, \tag{122}$$

and the first-best continuation action is

$$y^{\mathrm{FB}}(z) = \mathbf{1}\left\{\ell^{\mathrm{FB}}(c) \le z < \theta\right\}. \tag{123}$$

(ii) *Among complete pooling, regular binary action feedback, and the regular $L/M/H$ benchmark, every strictly informative policy is strictly dominated. An optimal mechanism can be chosen to induce*

$$x^* = 1, \qquad y^*(z) = 1 \quad \textit{for every } z < \theta, \tag{124}$$

and to send the same continuation message after every noncompletion realization.

(iii) *Contractual feedback is therefore strictly coarser than first-best feedback. The first best separates the two nonempty regions*

$$(-\infty, \ell^{\mathrm{FB}}(c)) \qquad \textit{and} \qquad [\ell^{\mathrm{FB}}(c), \theta), \tag{125}$$

whereas the optimal contract pools all of $(-\infty, \theta)$. In particular, the optimal contract induces excessive continuation effort for every

$$z < \ell^{\mathrm{FB}}(c). \tag{126}$$

(iv) *Uniformly on $\mathcal{K}$,*

$$\ell^{\mathrm{FB}}(c) = \theta - \mu + \Phi^{-1}(c) + o(1), \tag{127}$$

and

$$\Pr_1\left\{Z_1 < \ell^{\mathrm{FB}}(c)\right\} = o(c). \tag{128}$$

Hence the allocative surplus loss caused by contractual overwork is

$$\int_{-\infty}^{\ell^{\mathrm{FB}}(c)} g_1(z)\big[c - D(z)\big]\,dz = o(c^2). \tag{129}$$

Proof. We divide the proof into five steps.

Step 1: the first best. For every $(\mu, \theta) \in \mathcal{K}$,

$$D(\theta) = \Phi(\mu) - \frac{1}{2}. \tag{130}$$

This expression is bounded uniformly away from zero on $\mathcal{K}$. Therefore, for all sufficiently small c, part (i) of Proposition **??** applies and gives the unique lower cutoff in (**??**).

At $c = 0$, second-period work is efficient after every finite $z < \theta$. The gain from first-period work is then the difference between the completion probabilities generated by (W, W) and (S, W):

$$\Gamma^{\mathrm{FB}}(\mu, \theta, 0) = P_1 + A_1 - P_0 - A_0 > 0. \tag{131}$$

The difference is continuous and strictly positive on the compact set $\mathcal{K}$, and hence is bounded uniformly away from zero. Continuity in c implies

$$\Gamma^{\mathrm{FB}}(\mu, \theta, c) > c \tag{132}$$

for all sufficiently small c, uniformly on $\mathcal{K}$. First-period work is therefore uniquely first-best.

Because $\ell^{\mathrm{FB}}(c) \to -\infty$, the normal tail ratio gives, uniformly on $\mathcal{K}$,

$$\frac{p\big(\ell^{\mathrm{FB}}(c)\big)}{q\big(\ell^{\mathrm{FB}}(c)\big)} \longrightarrow 0. \tag{133}$$

Thus (**??**) implies

$$q\big(\ell^{\mathrm{FB}}(c)\big) = c\big[1 + o(1)\big], \tag{134}$$

which yields (**??**). It follows that

$$\begin{aligned}\Pr_1\left\{Z_1 < \ell^{\mathrm{FB}}(c)\right\} &= \Phi\left(\ell^{\mathrm{FB}}(c) - \mu\right) \\ &= \Phi\left(\Phi^{-1}(c) + \theta - 2\mu + o(1)\right).\end{aligned} \tag{135}$$

On $\mathcal{K}$, $\theta - 2\mu \leq -\underline{\mu}$. A second application of Mills' ratio therefore gives (**??**). Since $0 < c - D(z) < c$ below the first-best cutoff, (**??**) follows.

Step 2: complete pooling. Let

$$S_{11} := P_1 + A_1 \tag{136}$$

denote the gross completion probability under work in both periods. The formulas in Proposition **??** imply, uniformly on $\mathcal{K}$,

$$b_{WW}^{\mathrm{pool}} = O(c), \qquad R^{WW}\big(b_{WW}^{\mathrm{pool}}\big) = O(c), \tag{137}$$

and hence

$$V_{WW}^{\text{pool}} = S_{11} - O(c). \tag{138}$$

At $c = 0$, the other pooled allocations have gross completion probabilities

$$P_0 + Q_0, \qquad P_1 + Q_1, \qquad P_0 + A_0, \tag{139}$$

all strictly below S_{11}. The gaps are continuous and uniformly positive on $\mathcal{K}$. Therefore (W, W) is the unique optimal pooled allocation for all sufficiently small c.

Step 3: any competing informative policy must approach pooling. Consider a sequence of regular informative mechanisms with $c \downarrow 0$ whose values weakly exceed V_{WW}^{pool}. A first-period shirk branch cannot be competitive: even if it induced second-period work after every noncompletion realization, its gross completion probability would be $P_0 + A_0$, which remains uniformly below S_{11}. Hence every competitive sequence induces first-period work.

Transfers are nonnegative. In view of (**??**), the gross completion probability under a competitive mechanism must converge to S_{11}. Thus the continuation shirk region must have vanishing g_1-weighted productivity loss. Since D is uniformly bounded away from zero in a neighborhood of θ, any upper shirk tail must collapse to θ. On such a collapsing upper tail, the likelihood ratio is bounded above and below, and

$$D(z) = D(\theta) + o(1). \tag{140}$$

The gross loss from shirking there is therefore first order in its probability. By contrast, every possible saving in continuation-effort incentives or success-contingent transfers is $O(c)$ times that probability. Nor can an upper-tail wage improve the provision of first-period incentives: by the monotone likelihood ratio property, the immediate-success event $Z_1 \geq heta$ is a strictly more efficient reward event than any noncompletion tail below $heta$. Eliminating the upper shirk tail also weakly relaxes the associated reporting constraints. Consequently, an upper shirk tail is strictly dominated for all sufficiently small c.

It remains only to consider a lower shirk tail

$$L = (-\infty, \ell), \qquad W = [\ell, \theta), \tag{141}$$

with $\ell \to -\infty$. This covers regular one-sided binary feedback. After the upper tail has been eliminated, the same lower boundary is also the only potentially useful boundary in the regular $L/M/H$ benchmark.

Step 4: separating the lower tail is more expensive than pooling it. Any competitive sequence must have all on-path bonuses and the immediate-success reward of order $O(c)$. Otherwise it would incur a transfer bounded away from zero on an event whose probability is bounded away from zero, while (**??**) converges to S_{11}.

Let

$$d(\ell) := w_W - w_S \tag{142}$$

denote the reporting-wage difference at the lower boundary. The least common wage is zero on the first-period work branch. Boundary indifference gives

$$d(\ell) = (1 - b_W)q(\ell) - (1 - b_S)p(\ell). \tag{143}$$

Since $b_S, b_W = O(c)$ and $p(\ell)/q(\ell) \to 0$ uniformly,

$$d(\ell) = q(\ell)\big[1 + o(1)\big]. \tag{144}$$

Write

$$\pi_L^a := \Pr_a\{Z_1 < \ell\}. \tag{145}$$

Because $\theta \le \mu$ on $\mathcal{K}$,

$$\pi_L^0 = \Phi(\ell) \le \Phi(\ell + \mu - \theta) = q(\ell), \tag{146}$$

and because $\theta - 2\mu \le -\underline{\mu}$,

$$\pi_L^1 = \Phi(\ell - \mu) = o\big(q(\ell)\big) \tag{147}$$

uniformly on $\mathcal{K}$.

Now merge the lower-tail report into the work report. Apart from the reporting wage, every coefficient that changes in the fixed-partition incentive problem is a lower-tail probability, success moment, or productivity moment. The fixed-partition problems are finite linear programs on each continuation-action branch. Their value functions are therefore locally Lipschitz in these region moments. Since all competitive incentive payments are $O(c)$, the largest possible reduction in nonreporting incentive cost obtained by separating the lower tail is bounded by

$$O\left(c\pi_L^0 + c\pi_L^1\right) = o\big(q(\ell)\big). \tag{148}$$

The allocative effect of the separation is nonpositive for the principal's gross return, because it replaces work by shirk on L.

The reporting wage in (**??**), however, cannot help induce first-period work. Indeed, if $\pi_W^a := N_a - \pi_L^a$, then

$$\pi_W^0 - \pi_W^1 = \Delta P - \left(\pi_L^0 - \pi_L^1\right) > 0 \tag{149}$$

for all sufficiently negative ℓ. The wage is therefore more likely after a first-period deviation to shirking and weakens, rather than strengthens, first-period obedience. On path it is paid after the work report, whose probability converges to $N_1 > 0$. It therefore creates an on-path transfer equal to

$$N_1 q(\ell)\big[1 + o(1)\big], \tag{150}$$

and may additionally require a larger immediate-success reward. The cost in (**??**) strictly dominates (**??**). Hence every nonempty lower-tail separation is strictly worse than complete pooling for all sufficiently small c.

Step 5: conclusion. Steps 3 and 4 eliminate every strictly informative regular binary policy. They also eliminate both tail reports in the regular $L/M/H$ benchmark. Complete pooling with (W, W) is therefore optimal in the stated mechanism classes. Since $c > 0$, the cutoff $\ell^{\mathrm{FB}}(c)$ is finite, so the first-best shirk region is nonempty. Pooling all noncompletion outcomes is therefore a strict coarsening of the first-best continuation partition and induces excessive effort precisely below the first-best cutoff. □

Economic intuition: honest reporting is expensive. The force behind Proposition **??** is not a technological difficulty in inducing the agent to shirk after a sufficiently low realization. Conditional on a low report, the principal can induce shirking simply by attaching no continuation bonus to that report. The real difficulty is inducing the principal to send the low report honestly after observing a realization for which continuation work would raise the chance of completion.

To separate a low-effort region from the work region, the contract must make the principal indifferent at the reporting cutoff. Boundary indifference requires a reporting-payment wedge of the form

$$w_W - w_S = (1 - b_W)q(\ell) - (1 - b_S)p(\ell). \tag{151}$$

Thus, even though the low report is intended to reduce continuation effort, truthful reporting requires compensating the principal for forgoing the additional success probability generated by the work report. Limited liability prevents the contract from imposing a negative payment following a dishonest work report, so this compensation must be delivered through a positive payment attached to truthful reporting.

This reporting payment is costly for two related reasons. First, the wage needed to support the cutoff is attached to the work report and is therefore paid throughout the work region, rather than only at the realizations where the continuation action differs from the pooled policy. Second, continuation payments tend to weaken first-period incentives. A deviation to low first-period effort makes continuation relatively more likely, so a wage paid following noncompletion rewards the deviation disproportionately. Restoring first-period obedience can then require a larger immediate-success reward. Consequently, informative feedback creates both a direct reporting rent and an indirect increase in the cost of motivating initial effort.

The first best does not face either problem. It observes z and directly chooses whether continuation effort is socially worthwhile, so it can assign low effort below $\ell^{\mathrm{FB}}(c)$ without paying anyone to reveal that the state lies in the low region. The contractual problem must instead compare the allocative gain from tailoring continuation effort to z with the incentive cost of making the associated report credible. In the region identified by Proposition **??**, the reporting and first-period incentive costs dominate that allocative gain.

The optimal contract therefore suppresses information that would be useful under full commitment. It pools the low state with the ordinary continuation states and recommends work after every noncompletion realization. Coarser feedback and excessive continuation effort are not accidental by-products of the contract. They are substitutes for the principal's missing commitment to report honestly. The mechanism deliberately accepts a distortion in the continuation action in order to avoid the rents required to implement a finer, first-best feedback partition.

Remark 3 (Why the sufficient region is one-sided)**.** *The condition $\theta \leq \mu$ is not asserted to be sharp. It gives the simple uniform bound* (**??**)*. When $\theta > \mu$, an extreme lower-tail report can be substantially more likely following first-period shirking than the boundary probability $q(\ell)$ that determines its reporting wage. Pooling still appears numerically robust just above the diagonal, but extending Proposition* **??** *to a symmetric neighborhood of $\theta = \mu$ requires a sharper comparison of the first-period incentive effect of this off-path tail.*

8 Numerical evidence and conjectures

The numerical comparisons impose Assumption **??**. At each parameter point, complete pooling is solved analytically, binary action feedback is optimized over reporting cutoffs and the finite payment vertices identified above, and the unrestricted three-message problem is solved by outer cutoff search combined with exact fixed-cutoff linear programs.

A structured grid used

$$\mu \in \{0.5, 1, 2, 4\}, \tag{152}$$

$$\frac{\theta}{\mu} \in \{0.5, 0.8, 1, 1.25, 1.75, 2.5\}, \tag{153}$$

and four cost levels equal to

$$0.15, \quad 0.40, \quad 0.65, \quad 0.90 \tag{154}$$

times the largest cost consistent with $\Gamma^{\mathrm{FB}} > c$. This gives 96 parameter points.

Globally optimal mechanism	Number of cases
Complete pooling	87
Binary action feedback	9
A strict additional gain from a third message	0

At all nine feedback-winning points, the unrestricted three-message value and the binary action-feedback value agreed to numerical tolerance. All nine mechanisms induced first-period work. In every case the upper cutoff was numerically equal to θ, so the optimal feedback was one-sided:

$$S \quad \text{for} \quad z < \ell^*, \qquad W \quad \text{for} \quad \ell^* \leq z < \theta. \tag{155}$$

The lower cutoff was generally not equal to $\theta - \mu$. Thus the main gain relative to symmetric work bands comes from allowing the work region to tilt endogenously.

Across the full 96-point grid, the unrestricted three-message mechanism sometimes improved on binary feedback when each was evaluated in isolation. Whenever that improvement was economically material, however, complete pooling dominated both. Separate low- and high-tail reports therefore did not survive on the global upper envelope.

Conjecture 1 (Nonempty action-feedback region)**.** *There is a nonempty set of primitives satisfying Assumption* **??** *for which*

$$V^{\mathrm{WS}} > V^{\mathrm{pool}}. \tag{156}$$

Conjecture 2 (Redundancy of a third message)**.** *Whenever informative feedback is globally optimal,*

$$V^{\mathrm{LMH}} = V^{\mathrm{WS}}. \tag{157}$$

Equivalently, an optimal unrestricted three-message mechanism can be chosen with identical contracts in the two shirk tails.

Conjecture 3 (One-sided feedback under early work)**.** *Whenever binary feedback is globally optimal and induces first-period work, an optimal mechanism can be chosen with*

$$W \quad \Longleftrightarrow \quad \ell^* \leq z < \theta \tag{158}$$

for some lower cutoff $\ell^* < \theta$.

Conjecture 4 (Cost of noncommitted feedback)**.** *Complete pooling is optimal on a large region of the parameter space even under Assumption* **??***. The reason is not that interim information lacks allocative value, but that report-dependent continuation contracts must make the principal's message sequentially rational. When feedback is used, the optimal mechanism reveals only the desired continuation action.*

9 Conclusion

The model yields a sharp hierarchy of information structures. Complete pooling avoids all reporting costs but sacrifices state-contingent continuation effort. Binary action feedback preserves the allocative benefit of review while pooling states that lead to the same continuation action. Under normal noise, the principal's reporting payoff is single-peaked, so the work recommendation region is an interval and the fixed-interval payment problem is finite and piecewise linear. Unrestricted three-message feedback can be computed as a benchmark, but the numerical evidence indicates that a third message adds no value whenever informative feedback is globally optimal.

The low-cost result adds a distinct lesson. Optimal contractual feedback can be strictly coarser than first-best feedback even though the omitted information changes the efficient continuation action. At and immediately below the diagonal $\theta = \mu$, the first best stops effort in an extreme lower tail, but separating that tail requires a reporting wage whose cost dominates the vanishing allocative gain. The optimal contract therefore pools the tail and induces work everywhere. The remaining analytical targets are to extend this coarsening result above the diagonal, establish the third-message redundancy conjecture, and characterize the intermediate-cost boundary between pooling and action feedback.